Program:5 GRADLE

STEP 1:Create a directory in Devops called MyGradleApp , go to the directory and execute the command “gradle init –type java-application”

```
brijesh@Ubuntu:~$ cd Devops
brijesh@Ubuntu:~/Devops$ ls
MyMavenApp  MyMavenGuavaApp  MyMavenSeleniumApp01  MyMavenSeleniumApp02  MyMavenSeleniumApp03  MyMavenSeleniumFirefox  Mytoken
brijesh@Ubuntu:~/Devops$ mkdir MyGradleApp
brijesh@Ubuntu:~/Devops$ cd MyGradleApp
brijesh@Ubuntu:~/Devops/MyGradleApp$ gradle init --type java-application
Starting a Gradle Daemon (subsequent builds will be faster)

BUILD SUCCESSFUL in 11s
2 actionable tasks: 2 executed
brijesh@Ubuntu:~/Devops/MyGradleApp$
```

STEP 2:Display Tree using tree command

```
brijesh@Ubuntu:~/Devops/MyGradleApp$ tree
.
├── build.gradle
├── gradle
│   └── wrapper
│       ├── gradle-wrapper.jar
│       └── gradle-wrapper.properties
├── gradlew
├── gradlew.bat
├── settings.gradle
└── src
    ├── main
    │   └── java
    │       └── App.java
    └── test
        └── java
            └── AppTest.java

7 directories, 8 files
brijesh@Ubuntu:~/Devops/MyGradleApp$ ls
build.gradle  gradle  gradlew  gradlew.bat  settings.gradle  src
brijesh@Ubuntu:~/Devops/MyGradleApp$
```

STEP 3: Now build using the command: “gradle build”

```
brijesh@Ubuntu:~/Devops/MyGradleApp$ gradle build
Download https://jcenter.bintray.com/com/google/guava/guava/23.0/guava-23.0.pom
Download https://jcenter.bintray.com/com/google/guava/guava-parent/23.0/guava-parent-23.0.pom
Download https://jcenter.bintray.com/com/google/j2objc/j2objc-annotations/1.1/j2objc-annotations-1.1.pom
Download https://jcenter.bintray.com/com/google/code/findbugs/jsr305/1.3.9/jsr305-1.3.9.pom
Download https://jcenter.bintray.com/org/codehaus/mojo/animal-sniffer-annotations/1.14/animal-sniffer-annotations-1.14.pom
Download https://jcenter.bintray.com/com/google/errorprone/error_prone_annotations/2.0.18/error_prone_annotations-2.0.18.pom
Download https://jcenter.bintray.com/org/codehaus/mojo/animal-sniffer-parent/1.14/animal-sniffer-parent-1.14.pom
Download https://jcenter.bintray.com/com/google/errorprone/error_prone_parent/2.0.18/error_prone_parent-2.0.18.pom
Download https://jcenter.bintray.com/org/codehaus/mojo/mojo-parent/34/mojo-parent-34.pom
Download https://jcenter.bintray.com/org/codehaus/codehaus-parent/4/codehaus-parent-4.pom
Download https://jcenter.bintray.com/com/google/errorprone/error_prone_annotations/2.0.18/error_prone_annotations-2.0.18.jar
Download https://jcenter.bintray.com/com/google/guava/guava/23.0/guava-23.0.jar
Download https://jcenter.bintray.com/com/google/code/findbugs/jsr305/1.3.9/jsr305-1.3.9.jar
Download https://jcenter.bintray.com/com/google/j2objc/j2objc-annotations/1.1/j2objc-annotations-1.1.jar
Download https://jcenter.bintray.com/org/codehaus/mojo/animal-sniffer-annotations/1.14/animal-sniffer-annotations-1.14.jar

BUILD SUCCESSFUL in 21s
7 actionable tasks: 7 executed
```

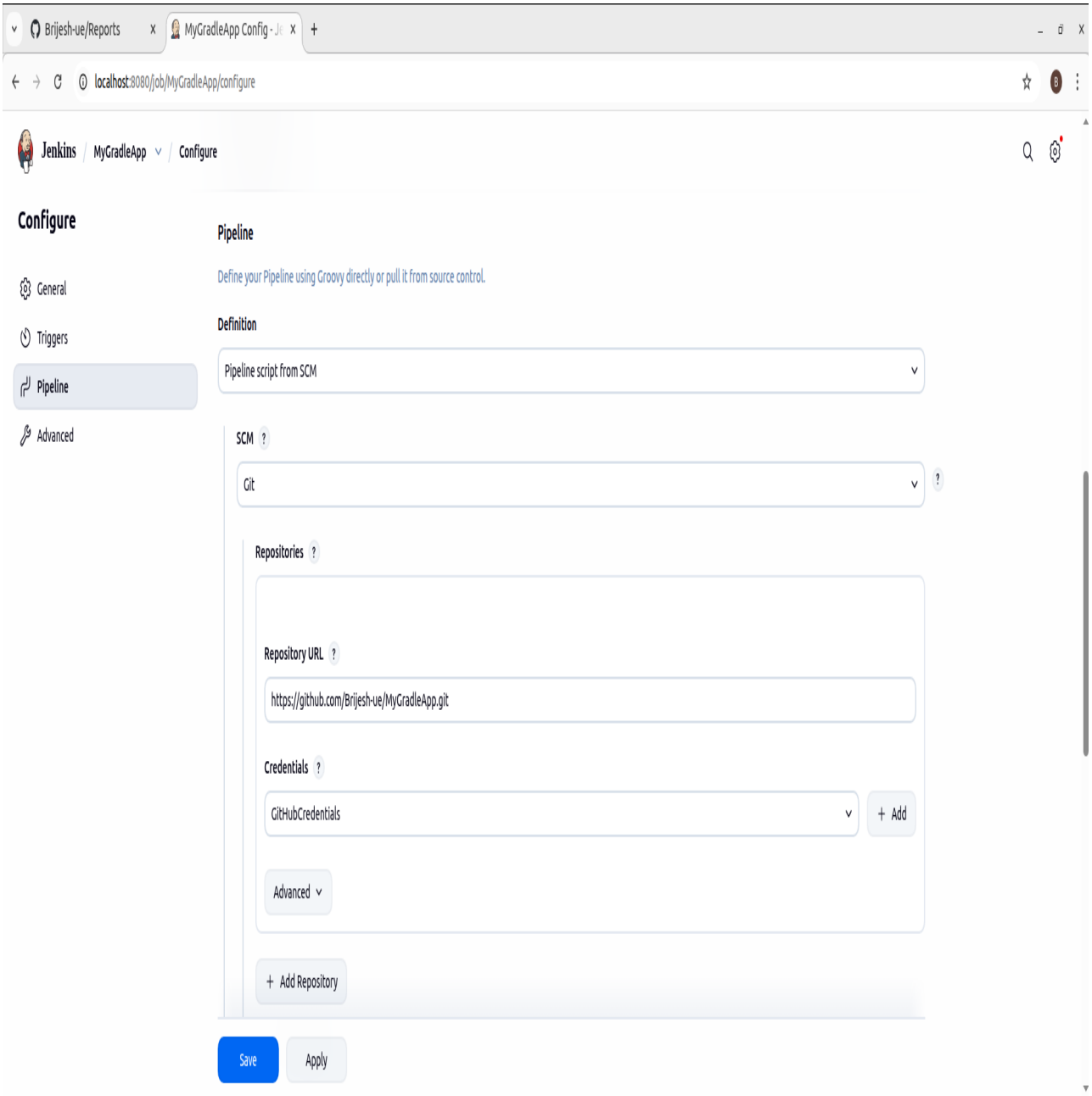
STEP 4: Execute “gradle run”
We get the output as follows

```
brijesh@Ubuntu:~/Devops/MyGradleApp$ gradle run

> Task :run
Hello world.


BUILD SUCCESSFUL in 2s
2 actionable tasks: 1 executed, 1 up-to-date
```

Gradle pipeline:

Step 1: Create a New Pipeline Job
Go to Dashboard → + New Item.
Enter name, select Pipeline, and click OK.

Step 2:Configure Pipeline from SCM
Scroll down and select Pipeline script from SCM.
SCM: Git Enter repository URL and select GitHub credentials.

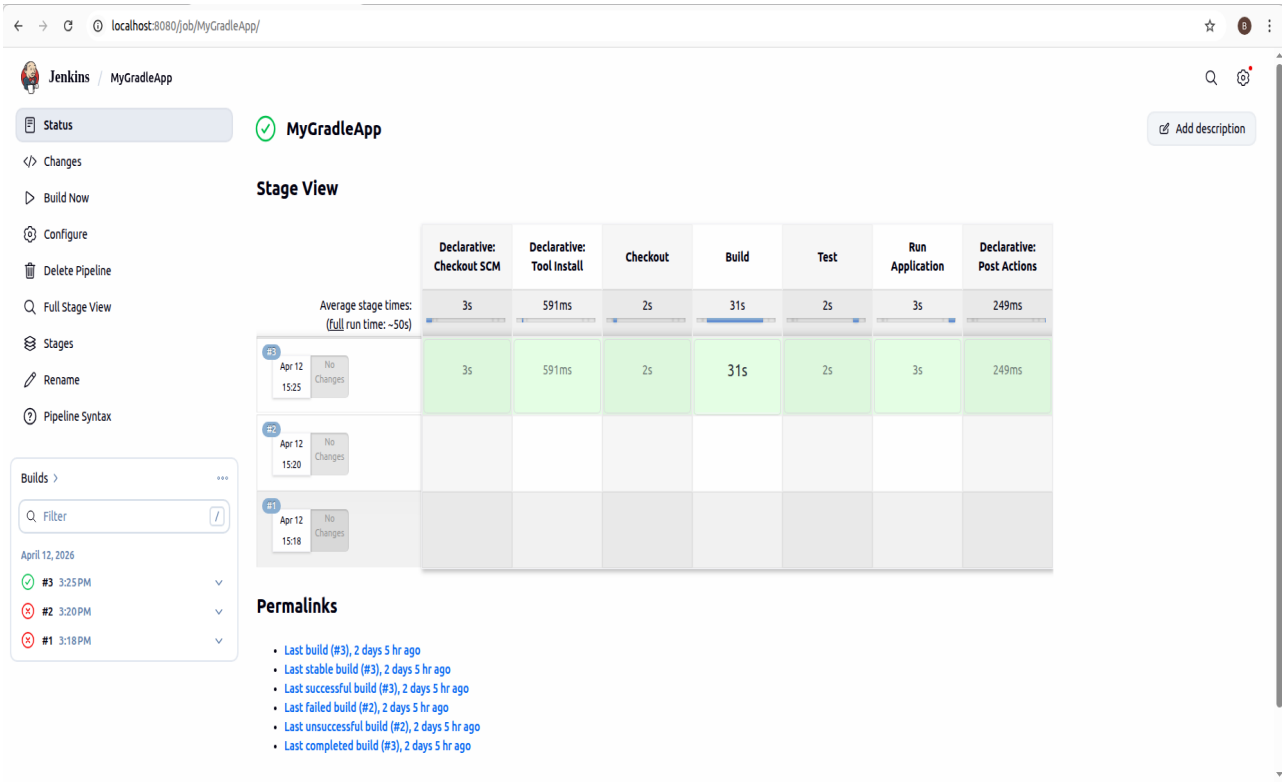


Step 3: Create Jenkinsfile
Create a Jenkinsfile in your MyGradleApp project and add the pipeline script.

```
pipeline {
    agent any  // Use any available agent

    tools {
        gradle 'Gradle'  // Ensure this matches the name configured in Jenkins
        jdk 'JDK'
    }
    stages {
        stage('Checkout') {
            steps {
                git branch: 'main', url:
'https://github.com/Brijesh-ue/MyGradleApp.git'
            }
        }

        stage('Build') {
            steps {
                sh 'gradle build'  // Run Gradle build
            }
        }

        stage('Test') {
            steps {
                sh 'gradle test'  // Run unit tests
            }
        }




        stage('Run Application') {
            steps {
                // Start the JAR application
                sh 'gradle run'
```

```
            }
        }


    }

    post {
        success {
            echo 'Build and deployment successful!'
        }
        failure {
            echo 'Build failed!'
        }
    }
}
```

Step 4: Build the Pipeline

Go to Jenkins and click Build Now.

Monitor Build Stages Use Stage View and Full Stage View to monitor execution.